

DAILY QUESTIONS

Basic Mathematics Practice

B.Sc (Hons.) Agriculture / B.Tech Biotechnology

Q1. Simplify:

$$\sqrt{98}$$

Q2. Rationalize:

$$\frac{1}{\sqrt{5}}$$

Q3. Solve:

$$|2x - 3| = 7$$

Q4. Find:

$$(3 + 2i)(4 - i)$$

Q5. Prove that:

$$\sqrt{3}$$

is irrational.

Q1. Let:

$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 4, 5, 6\}$$

Find:

(a) $A \cup B$

(b) $A \cap B$

(c) $A - B$

Q2. Verify:

$$(A \cup B)' = A' \cap B'$$

Q3. Draw Venn diagram for:

$$A \cap B$$

Q1. Find domain:

$$f(x) = \frac{1}{x^2 - 9}$$

Q2. Evaluate:

$$f(x) = x^2 + 3x + 2$$

Find:

$$f(2)$$

Q3. Determine whether:

$$f(x) = x^2$$

is one-one.

Q4. Sketch graph:

$$y = |x|$$

Q1. Evaluate:

$$\lim_{x \rightarrow 2} (x^2 + 3x)$$

Q2. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

Q3. Check continuity:

$$f(x) = \begin{cases} x + 1, & x < 2 \\ 2x, & x \geq 2 \end{cases}$$

Q4. Define removable discontinuity.